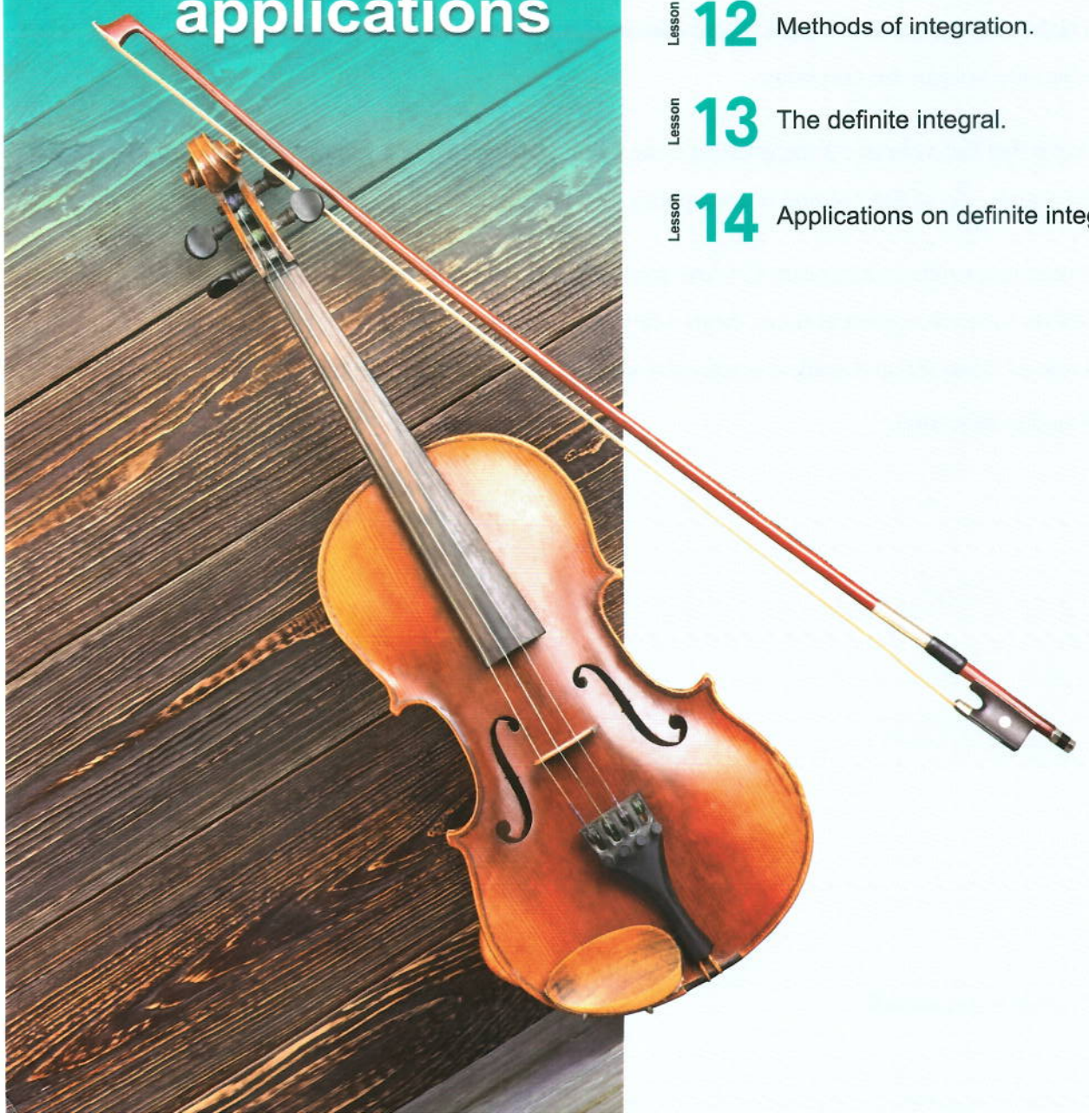
UNIT

# THREE

# The definite integral and its applications



⊙ **Exercise on pre-requirements for unit 3**

Lesson **11** Integral of exponential and logarithmic functions.

Lesson **12** Methods of integration.

Lesson **13** The definite integral.

Lesson **14** Applications on definite integral.

# Exercise on Pre-requirements for unit 3
# Introduction for integral - differential

From the school book | Understand | Apply | Higher Order Thinking Skills

**1** **Find the differential for each of the following :**

( 1 ) $y = 3x^2 - 1$

( 2 ) $y = (2x + 5)^4$

( 3 ) $y = e^{2x-3}$

( 4 ) $y = \frac{x}{x-1}$

( 5 ) $v = \frac{4}{3}\pi r^2$

( 6 ) $y = \sqrt{(2x+3)^5}$

( 7 ) $y = \left(x + \frac{1}{x}\right)^3$

( 8 ) $y = e^{x^2+3x}$

( 9 ) $y = e^{\cos 2x}$

(10) $z = \ln(x^2 + 1)$

(11) $z = \left(\ln \frac{10}{x}\right)^2$

(12) $z = \ln \cot(3x - 1)$

**2** **If each of $z$ , $\ell$ is a function in $x$ , then find the differential for each of the following :** ( 1 ) $y = z \cdot \ell$ ( 2 ) $y = \frac{z}{\ell}$

**3** **If $x^2 + y^2 = 25$ , then find dy interms of $x$ , $y$ , $dx$**

# Exercise 11

## On integral of exponential and logarithmic functions

From the school book | Understand | Apply | Higher Order Thinking Skills

**1** Find each of the following :

(1) $\int e^{4x}\,dx$

(2) $\int e^{-x}\,dx$

(3) $\int e^{\frac{-3}{4}y}\,dy$

(4) $\int -6\,e^{0.2y}\,dy$

(5) $\int 3\sqrt{2}\,e^{-\sqrt{2}n}\,dn$

(6) $\int -e^{-5z}\,dz$

(7) $\int \pi\,e^{x}\,dx$

(8) $\int e^{1-3x}\,dx$

(9) $\int 5^{x}\,dx$

(10) $\int 2^{-5x}\,dx$

**2** Find each of the following :

(1) $\int (4x - 2e^{x})\,dx$

(2) $\int (3x^{2} - 4e^{2x})\,dx$

(3) $\int (1 + 4e^{-3e})\,dx$

(4) $\int (e^{2k} + 4k)\,dk$

(5) $\int (2x^{2} + \sin 2x - 4e^{3x})\,dx$

(6) $\int \left(\frac{e^{x} + e^{-x}}{2}\right) dx$

(7) $\int \left(\frac{e^{x} + 3}{e^{x}}\right) dx$

(8) $\int \frac{e^{x} + e^{-x}}{e^{x}}\,dx$

(9) $\int \left(\frac{e^{3x} + 2e^{2x} + 4}{e^{x}}\right) dx$

(10) $\int \left(e^{x} + \frac{1}{e^{x}}\right)^{2} dx$

(11) $\int 2e^{x}(e^{x} + 1)^{2}\,dx$

(12) $\int \left(\frac{e^{2x} - 2e^{x} - 8}{e^{x} - 4}\right) dx$

(13) $\int (x^{2e} + e^{3x})\,dx$

(14) $\int (\pi^{4} + \pi^{x})\,dx$

**3** Find each of the following :

(1) $\int 4x\,e^{x^{2}+1}\,dx$

(2) $\int x^{2}\,e^{x^{3}+1}\,dx$

(3) $\int (x-3)\,e^{x^{2}-6x-5}\,dx$

(4) $\int x^{2}(4x-3)\,e^{x^{4}-x^{3}}\,dx$

(5) $\int (1 + \sin x)\,e^{(x-\cos x)}\,dx$

(6) $\int \sin x \times e^{\cos x}\,dx$

(7) $\int (\cos x\,e^{\sin x} + 3x^{2})\,dx$

(8) $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}\,dx$

(9) $\int \frac{dx}{\sqrt{x}\, e^{\sqrt{x}}}$

(10) $\int \frac{e^x}{\sqrt{1+2e^x}}\, dx$

(11) $\int 9\,(e^x+1)^8\, e^x\, dx$

(12) $\int 7^{\cos x} \sin x\, dx$

**4** **Find each of the following :**

(1) $\int \frac{2}{x}\, dx$

(2) $\int \frac{1}{x+1}\, dx$

(3) $\int \left(x+\frac{1}{x}\right) dx$

(4) $\int \left(\frac{3}{x}+4\,e^{-2x}\right) dx$

(5) $\int \frac{4}{3x \ln 5}\, dx$

(6) $\int \frac{\ln 3}{x}\, dx$

(7) $\int \frac{\ln x^2}{x \ln x^3}\, dx$

(8) $\int \frac{\ln x}{x \ln \sqrt{x}}\, dx$

(9) $\int \left(\frac{2e}{x}+\frac{x}{e}\right) dx$

(10) $\int \left(\frac{e^3}{x}+x \ln 3\right) dx$

(11) $\int \left(3\,e^{-x}+\frac{\pi}{x}\right) dx$

(12) $\int \left(3x^2+\frac{5}{x}\right) dx$

(13) $\int \left(\sin x-\frac{2}{x}+e^x\right) dx$

(14) $\int \left(e^x+3^x+\frac{1}{x}\right) dx$

(15) $\int \left(\frac{5x+7}{x}+e^x\right) dx$

(16) $\int \left(\frac{6x^2-5}{3x}\right) dx$

(17) $\int \frac{x^2-4}{x^2-2x}\, dx$

(18) $\int \left(\sqrt{x}-\frac{3}{\sqrt{x}}\right)^2 dx$

(19) $\int \frac{(2x-1)^2}{3x}\, dx$

(20) $\int \frac{4\,e^x+x\,e^{2x}}{x\,e^x}\, dx$

**5** **Find each of the following :**

(1) $\int \frac{dx}{4x-1}$

(2) $\int \frac{3x}{x^2-1}\, dx$

(3) $\int \frac{9+6x}{x^2+3x}\, dx$

(4) $\int \frac{x+1}{x^2+2x+7}\, dx$

(5) $\int \frac{3x^2-5}{x^3-5x+1}\, dx$

(6) $\int \frac{x^2+2}{x^3+6x+1}\, dx$

(7) $\int \frac{2x-1}{x-1}\, dx$

(8) $\int \frac{x^2}{x+1}\, dx$

(9) $\int \frac{e^x}{e^x+3}\, dx$

(10) $\int \frac{e^{2x}+1}{e^{2x}-1}\, dx$

## 6 Find each of the following :

(1) $\int \cot x \, dx$

(2) $\int \sec x \, dx$

(3) $\int \frac{\sin x}{1 + \cos x} \, dx$

(4) $\int \frac{\sec^2 x}{\tan x} \, dx$

(5) $\int \frac{\csc^2 x}{\cot x} \, dx$

(6) $\int \frac{\sin x + \cos x}{\sin x - \cos x} \, dx$

(7) $\int \tan (3x + 1) \, dx$

(8) $\int \sec (5 - 2x) \, dx$

(9) $\int \csc (4x + 7) \, dx$

(10) $\int \frac{1 - \tan x}{1 + \tan x} \, dx$

(11) $\int \frac{\sec x \, . \tan x}{\sec x - 1} \, dx$

(12) $\int \frac{\sin 2x + 2 \cos x}{\sin^2 x + 2 \sin x - 1} \, dx$

(13) $\int \tan 6x \, dx$

(14) $\int \cot^3 x \, dx$

(15) $\int \frac{1 + \tan x}{x + \ln (\sec x)} \, dx$

(16) $\int \frac{\sin 2x}{\sin 4x} \, dx$

(17) $\int \frac{\sin 2x}{1 + \sin^2 x} \, dx$

(18) $\int \frac{x^{e-1} + e^{x-1}}{x^e + e^x} \, dx$

## 7 Choose the correct answer from the given ones :

(1) *(1st Session 2023)* $\int \frac{1}{2x + e} \, dx =$ ..... + c

(a) $2 \ln |x + e|$ (b) $2 \ln |2x + e|$

(c) $\frac{1}{2} \ln |x + e|$ (d) $\frac{1}{2} \ln |2x + e|$

(2) *(2nd Session 2023)* $\int \frac{3x^2}{x^3 + 4} \, dx =$ ..... + c

(a) $x^3 \ln |x^3 + 4|$ (b) $3 \ln |x^3 + 4|$ (c) $6 \ln |x^3 + 4|$ (d) $\ln |x^3 + 4|$

(3) $\int 4x e^{x^2} \, dx =$ ..... + c

(a) $\frac{1}{2} e^{x^2}$ (b) $e^{x^2}$ (c) $2 e^{x^2}$ (d) $4 e^{x^2}$

(4) *(1st Session 2018)* $\int \tan \theta \, d\theta =$ ..... + c

(a) $- \ln |\cos \theta|$ (b) $- \ln \cos \theta$ (c) $\ln \cos \theta$ (d) $|\ln \cos \theta|$

(5) *(1st Session 2017)* $\int \frac{\ln x^2}{\ln x} \, dx =$ ..... + c

(a) $\frac{x}{2}$ (b) $\frac{1}{x}$ (c) $2x$ (d) $\ln |x|$

( 6 ) *(1st Session 2024)* $\int \frac{1}{x} \ln \frac{1}{x} \, dx =$ ..... + c

(a) $\frac{1}{2} (\ln x)^2$ (b) $-2 (\ln x)^2$ (c) $2 (\ln x)^2$ (d) $-\frac{1}{2} (\ln x)^2$

( 7 ) $\int \frac{1}{x \ln x^3} \, dx =$ ..... + c

(a) $3 \ln |x|$ (b) $3 \ln |\ln x|$ (c) $\frac{1}{3} \ln |x|$ (d) $\frac{1}{3} \ln |\ln x|$

( 8 ) $\int e^2 \, dx =$ ..... + c

(a) $e^2 x$ (b) $\frac{1}{3} e^3$ (c) $e^2$ (d) $\frac{1}{3} e^{3x}$

( 9 ) $\int e^{\ln x} \, dx =$ ..... + c

(a) $\frac{x^2}{2} \ln x$ (b) $\ln x$ (c) $\frac{1}{2} x^2$ (d) $e^x$

(10) *(2nd Session 2021)* $\int 4 \sin 2x \, e^{\cos 2x} \, dx =$ ..... + c , where c is constant.

(a) $-2 e^{\cos 2x}$ (b) $2 e^{\cos 2x}$ (c) $4 e^{\cos 2x}$ (d) $-4 e^{\cos 2x}$

(11) *(1st Session 2022)* $\int \left(3x^2 - \frac{3}{x}\right) dx =$ ..... + c where $x \neq 0$

(a) $x^3 - \ln \left|\frac{3}{x}\right|$ (b) $3x^3 - 3 \ln |x|$ (c) $x^3 - 3 \ln |x|$ (d) $3x^3 - \ln \left|\frac{3}{x}\right|$

(12) *(Speciman 2025)* $\int \frac{2 \cos x - 3 \sin x}{6 \cos x + 4 \sin x} \, dx =$ ..... + c

(a) $\ln |6 \cos x + 4 \sin x|$ (b) $2 \ln |6 \cos x + 4 \sin x|$

(c) $\frac{1}{2} \ln |3 \cos x + 2 \sin x|$ (d) $\frac{1}{2} \ln |2 \cos x + 3 \sin x|$

(13) *(Speciman 2025)* $\int \frac{(x+2)}{x^2 + 4x - 5} \, dx =$ .....

(a) $\ln |x^2 + 4x - 5| + c$ (b) $2 \ln |x^2 + 4x - 5| + c$

(c) $\frac{1}{2} \ln |x^2 + 4x - 5| + c$ (d) $\frac{1}{2} \ln |x + 2| + c$

(14) *(Speciman 2025)* $\int \frac{7x^2}{5 - 4x^3} \, dx =$ .....

(a) $\frac{7}{12} \ln |4x^3 - 5| + c$ (b) $\frac{12}{7} \ln |4x^3 - 5| + c$

(c) $\frac{-7}{12} \ln |4x^3 - 5| + c$ (d) $\frac{-12}{7} \ln |4x^3 - 5| + c$

(15) *(2nd Session 2022)* $\int \frac{\sec x \tan x}{3 + \sec x} \, dx =$ ..... + c

(a) $\ln |\sec x \tan x|$ (b) $\ln \left|\frac{\tan x}{3}\right|$ (c) $\ln |\tan x|$ (d) $\ln |3 + \sec x|$

**(16)** $\int (1 + \tan^2 x)\, e^{1 + \tan x}\, dx =$ ..... $+ c$

(a) $e^{\cot x}$ (b) $e^{1 + \tan x}$ (c) $e^{1 + \cot x}$ (d) $e^{\sec^2 x}$

**(17)** *(2nd Session 2018)* $\int \frac{x + 2}{x + 1}\, dx =$ ..... $+ c$

(a) $1 + \ln (x + 1)$ (b) $x - \ln |x + 1|$

(c) $x + \ln |x + 2|$ (d) $x + \ln |x + 1|$

**(18)** *(Speciman 2025)* $\int \frac{dx}{e^{-x} + 1} =$ .....

(a) $\ln (e^{-x} + 1) + c$ (b) $x + c$

(c) $\ln (e^{x} + 1) + c$ (d) $- \ln (e^{-x} + 1) + c$

**(19)** *(Speciman 2025)* $\int \frac{e^{2x} - 5e^{x} + 6}{e^{x} - 3}\, dx =$ ..... $+ c$

(a) $e^{-x} - 2x$ (b) $e^{x} - 2x$

(c) $e^{x} + 2x$ (d) $2x - e^{x}$

**(20)** $\int \frac{e^{3x} - 1}{e^{x} - 1}\, dx =$ ..... $+ c$

(a) $\frac{1}{2} e^{2x} + e^{x} + x$ (b) $e^{2x} + e^{x} + 1$

(c) $e^{2x} - e^{x} + 1$ (d) $\frac{1}{3} e^{3x} - x$

**(21)** If $\int e^{f(x)}\, dx = \frac{1}{2} x^2 + c$ , then $f(x)$ could be .....

(a) $x$ (b) $\log x$ (c) $\ln x$ (d) $x^2$

**(22)** $\int \frac{\ln x^2}{x}\, dx =$ ..... $+ c$ where $x > 0$

(a) $\ln x$ (b) $\ln |\ln x|$ (c) $\frac{1}{2} (\ln x)^2$ (d) $(\ln x)^2$

**(23)** $\int \left( \sum_{n=0}^{\infty} \frac{x^n}{\lfloor n} \right) dx =$ ..... $+ c$

(a) $e^{x}$ (b) $e\,x$ (c) $x^{2e}$ (d) $e^{x^2}$

**(24)** $\int \frac{1 + \ln x}{8 + x \ln x}\, dx =$ ..... $+ c$

(a) $\ln |x \ln x|$ (b) $\ln |8 + x \ln x|$ (c) $8 + \ln |x|$ (d) $\ln 8 + \ln x$

**(25)** $\int \frac{6}{x + x \ln x}\, dx =$ ..... $+ c$

(a) $6 (1 + \ln x)$ (b) $6 \ln |1 + \ln x|$ (c) $\frac{6}{\ln |1 + \ln x|}$ (d) $\frac{6}{\ln |x + \ln x|}$

**(26)** $\int (7^{2\log_7 \cos x} - 5^{2\log_5 \sin x})\, dx = \ldots\ldots\ldots\ldots + c$

(a) $x$ (b) $\cos x - \sin x$ (c) $\frac{1}{2}\sin 2x$ (d) $-\frac{1}{2}\cos 2x$

**(27)** *(Trial 2021)* If $f(x) = \int \frac{1-(\ln x)^2}{x}\, dx$ where $f(1) = 0$ , then $f(e) = \ldots\ldots\ldots\ldots$

(a) $-\frac{2}{3}$ (b) $-\frac{1}{3}$ (c) $\frac{1}{3}$ (d) $\frac{2}{3}$

**(28)** *(1st Session 2021)* $\int \frac{x}{x^2-2x+1}\, dx = \ldots\ldots\ldots\ldots + c$ where c is constant.

(a) $\ln|x-1| - \frac{1}{x-1}$ (b) $\ln|x-1| + \frac{1}{x-1}$

(c) $\ln|x-1| + (x-1)^2$ (d) $\ln|x^2-2x+1| + \frac{1}{x-1}$

**(29)** *(1st Session 2022)* $\int \frac{2}{(2\cos^2 x - 1)^2 (1+\tan 2x)}\, dx = \ldots\ldots\ldots\ldots + c$

(a) $\ln|1+\sec 2x|$ (b) $\ln|1+\tan 2x|$

(c) $\ln|\cos 2x| + \ln|1+\tan 2x|$ (d) $\ln|\cos 2x - \tan 2x|$

**(30)** *(2nd Session 2024)* $\int \sin x \sec x \ln \sec x\, dx = \ldots\ldots\ldots\ldots + c$

(a) $2\ln \sec^2 x$ (b) $\frac{1}{2}(\ln \sec^2 x)$ (c) $\frac{1}{2}(\ln \sec x)^2$ (d) $2\ln \sec x$

**(31)** *(2nd Session 2023)* $\int \frac{1-2\sin^2 x}{2\sin x \cos x}\, dx = \ldots\ldots\ldots\ldots + c$

(a) $\frac{-1}{2}\ln|\cos 2x|$ (b) $2\ln|\sin 2x|$ (c) $\frac{1}{2}\ln|\sin 2x|$ (d) $-2\ln|\cos 2x|$

**(32)** $\int \frac{2\tan x}{1-\tan^2 x}\, dx = \ldots\ldots\ldots\ldots + c$

(a) $\frac{1}{2}\tan 2x$ (b) $2\sec^2 2x$

(c) $-\ln|\cos x|$ (d) $-\frac{1}{2}\ln|\cos 2x|$

**(33)** $\int (\cot^4 x + \cot^2 x)\, dx = \ldots\ldots\ldots\ldots + c$

(a) $\frac{1}{3}\cot^3 x$ (b) $\tan x$ (c) $\ln|\sin^2 x|$ (d) $-\frac{1}{3}\cot^3 x$

**(34)** *(1st Session 2025)* If $f'''(x) = \frac{1}{2}\left[e^x + \frac{1}{e^x}\right]$ , $f(0) = 1$ , $f'(0) = 0$ , then $f(x) = \ldots\ldots\ldots\ldots$

(a) $-f'(x)$ (b) $f'(x)$ (c) $-f'''(x)$ (d) $f'''(x)$

**(35)** $\int \frac{\sin(\ln x)}{x}\, dx = \ldots\ldots\ldots\ldots$

(a) $-\cos x(\ln x) + c$ (b) $\cos(\ln x) + c$

(c) $-\cos(\ln x) + c$ (d) $\frac{1}{2}[\sin(\ln x)]^2 + c$

**(36)** If $\sin^2 x \, dy = y \, dx$ , then .....

(a) $|y| = e^{-\cot x + c}$ (b) $|y| = e^{\csc x + c}$

(c) $|y| = e^{\cot x + c}$ (d) $|y| = e^{-\csc x + c}$

**(37)** ***(1st Session 2024)*** If the slope of the tangent to the curve $y = f(x)$ at any point $(x, y)$ on it equals $2 e^{2x-1}$ and $f(1) = 4e$ , then $f\left(\frac{1}{2}\right)$ = .....

(a) $1 + 3e$ (b) $3e$ (c) $2e$ (d) $1 - e$

**(38)** ***(2nd Session 2024)*** If the slope of the tangent to the curve $y = f(x)$ at any point $(x, y)$ lying on it equals $2x - 2$ and $f(1) = 2$ , then $f(0)$ = .....

(a) 3 (b) 1 (c) $-1$ (d) $-2$

**(39)** If the slope of the tangent to the curve at any point $(x, y)$ on it equals $4 e^{2x}$ , $f(0) = 2$ , then : $f(-2)$ = .....

(a) 4 (b) $4 e^{-4}$ (c) $2 e^{-4}$ (d) $2e$

**(40)** ***(1st Session 2021)*** If $P = f(n)$ where $P, n \in \mathbb{R}^+$ and the rate of change of P with respect to n varies inversely with n , $f(1) = 200$ and $f\left(\sqrt[3]{e}\right) = \frac{700}{3}$ , then $f(3)$ = .....

(a) $100 \ln(3e^2)$ (b) $200 \ln(9e)$ (c) $100 \ln(e^2 + 3)$ (d) $200 \ln(9 + e)$

**(41)** ***(1st Session 2022)*** If the slope of the tangent to the curve of the function $y = f(x)$ at any point $(x, y)$ which lies on it is $e^{\cos x} \sin x$ and $f(0) = e$ , then $f\left(\frac{\pi}{2}\right)$ = .....

(a) $2e - 1$ (b) $e - 1$ (c) $1 - e$ (d) $1 - 2e$

**(42)** ***(1st Session 2021)*** If $x \frac{dy}{dx} - y = 3$ and $x = 1$ when $y = -2$ , then the relation between $x$ , y is .....

(a) $|y| = |x + 3|$ (b) $|y + 3| = |x|$ (c) $y + x = -1$ (d) $y + 2x = 0$

**(43)** ***(2nd Session 2021)*** If $\int \frac{2}{y} dy = \int \frac{1}{x} dx$ , then $\ln y^2$ = ..... + c where c is constant.

(a) $\ln|x|$ (b) $|x|$ (c) $\ln|x + y|$ (d) $\ln|x - y|$

**(44)** ***(2nd Session 2023)*** If the slope of the tangent to the curve of a function : $y = f(x)$ at any point $(x , y)$ lies on it equals $(-6 \cos^5 x \sin x)$ and the curve is passing through the point $\left(\frac{\pi}{2} , 0\right)$ , then y = .....

(a) $-\cos^6 x$ (b) $\cos^6 x$ (c) $\frac{1}{6}\cos^6 x$ (d) $-\frac{1}{6}\cos^6 x$

**(45)** ***(1st Session 2023)*** If the slope of the normal to the curve of the function : $y = f(x)$ at any point $(x , y)$ lies on it equals $(\sec x \csc x)$ and the curve passes through the point $\left(\frac{\pi}{6} , \frac{-1}{8}\right)$ , then the curve cuts y-axis at the point .....

(a) $\left(0 , \frac{-1}{4}\right)$ (b) $\left(0 , \frac{1}{4}\right)$ (c) $\left(0 , \frac{1}{2}\right)$ (d) $(0 , 0)$

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**8** If the slope of the tangent to the curve of the function at any point $(x , y)$ on it equals $7 - 2e^x$ and $f(\ln 2) = 3$ , find $f(x)$

« $f(x) = 7x - 2e^x + 7 \ln \frac{e}{2}$ »

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**9** 📖 If the slope of the tangent to the curve of the function $f$ at any point $(x , y)$ equals $2e^{-\frac{1}{2}x}$ and $f(0) = 1$ , find $f(3)$

« $5 - 4e^{-\frac{3}{2}}$ »

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**10** 📖 The slope of the tangent to the curve of the function $f$ at any point $(x , y)$ on it equals $\frac{1}{2x - e}$ and $f(e) = \frac{1}{2}$ , find $f(2e)$

« $\frac{1}{2}(1 + \ln 3)$ »

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**11** 📖 The slope of the tangent to a curve at any point $(x , y)$ on it equals $\frac{3x + 2}{x}$ , find the equation of the curve if known that it passes through point $(e , 3e + 5)$

« $y = 3x + 2 \ln|x| + 3$ »

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**12** 📖 If the rate of change in the surface area of a lamina A (in square centimeters) with respect to time t (second) is identified by the relation $\frac{dA}{dt} = e^{-0.1t}$ and the area of the lamina at the beginning of the change equals 80 $cm^2$ , find the surface area of the lamina after 10 second.

« $(90 - 10e^{-1})$ $cm^2$ »

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**13** 📖 If the rate of change of the sales in a factory is inversely proportional to time in weeks and the sales of the factory after 2 weeks and 4 weeks 200 units and 300 units respectively , find the sales of the factory after 8 weeks.

« 400 units »

**14** If the slope of the tangent at any point $(X, y)$ on the curve of the function $f$ is inversely proportional to $X$ and the slope of the tangent equals 2 when $X = 4$ and $y = 2$, find y in terms of $X$.

**15** **Find each of the following :**

(1) $\int \frac{1}{X \ln X} \, dX$

(2) $\int \frac{4}{X \ln 3X} \, dX$

(3) $\int \frac{(\ln X)^2}{X} \, dX$

(4) $\int \frac{(\log X)^2}{X} \, dX$

(5) $\int \frac{(1 + \ln X)^2}{X} \, dX$

(6) $\int \frac{2X}{(X+1)^2} \, dX$

(7) $\int \frac{\sqrt{1 + \ln X}}{X} \, dX$

(8) $\int \frac{X+1}{X\sqrt{X + \ln X}} \, dX$

## Higher skills

**16** **Choose the correct answer from those given :**

(1) If $\int \frac{f(X)}{X} \, dX = \ln|X| + X^2 + c$, then $f(X) =$ .....

(a) $X^2 + X + 1$ (b) $X^3 + X^2 + 1$ (c) $2X^2 + 1$ (d) $X^2 + 1$

(2) $\int \frac{1}{X^3} (\ln X^X)^2 \, dX =$ ..... $+ c$

(a) $\frac{X^3}{3} (\ln X) + X$ (b) $\frac{1}{3} (\ln X)^3$ (c) $3 \ln|\ln X|$ (d) $\frac{1}{X} (\ln X)^2$

(3) $\int \frac{\tan X}{1 - \cos X} \, dX =$ ..... $+ c$

(a) $\ln|1 - \tan X|$ (b) $\ln|1 - \cos X|$ (c) $\ln|\sec X - 1|$ (d) $\log|\sec X - 1|$

**17** **Find each of the following :**

(1) $\int \frac{1}{X} \cos(\ln X) \, dX$

(2) $\int a^{3 \log_a X} \, dX$

(3) $\int \left(\frac{3}{X} - \frac{2X-1}{X^2 - X + 2}\right) dX$

(4) $\int \frac{4X^2 + X + 1}{X^3 - 1} \, dX$

(5) $\int \frac{1}{X} \ln(\sin(\ln X)) \cot(\ln X) \, dX$

(6) $\int (X^3 + a^X + e^X . a^X + \sin a) \, dX$

(7) $\int (e^{X \ln a} + e^{a \ln X} + e^{a \log a}) \, dX$